

# Week 8: Short Review Test

## Weeks 1–7 Concepts

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### Instructions

- Show all work for full credit
- Circle your final answers
- No calculators

## Part A: Foundations & Equations

### Problem 1

Simplify:  $6 + 15 \div 3 - 2 \times 4$

### Problem 2

Simplify:  $5(3x - 2) = -2(x + 4)$

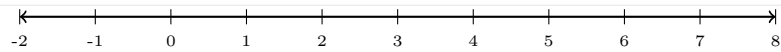
### Problem 3

Solve for  $W$ :  $P = 2L + 2W$

## Part B: Inequalities

### Problem 4

Solve and graph:  $-3x + 5 \leq -4$



Write in interval notation: \_\_\_\_\_

### Problem 5

Solve:  $|2x - 3| < 7$

Write in interval notation: \_\_\_\_\_

## Part C: Linear Functions & Lines

### Problem 6

Find the slope of the line through  $(2, 5)$  and  $(6, -3)$ .

### Problem 7

Write the equation of a line with slope  $\frac{3}{4}$  and y-intercept  $-2$ .

### Problem 8

Find the equation of the line through  $(3, -1)$  and perpendicular to  $y = -2x + 5$ .

## Part D: Systems of Equations

### Problem 9

Solve the system:

$$y = 3x - 4$$

$$y = -x + 8$$

## Part E: Applications

### Problem 10

A gym membership costs \$50 per month plus a one-time \$100 registration fee.

Write an equation for the total cost  $C$  after  $m$  months.

Find the total cost after 6 months.

### Problem 11

A store sells notebooks and pens.

There were 80 items sold in total.

Notebooks cost \$3 each and pens cost \$2 each.

The total revenue was \$190.

How many notebooks were sold?

**Problem 12**

Two cars leave from the same location at the same time, traveling in opposite directions.

Car A travels at 55 mph and Car B travels at 65 mph.

After how many hours will they be 300 miles apart?

## Answer Key

### Problem 1

$$6 + 15 \div 3 - 2 \times 4 = 6 + 5 - 8 = 3$$

### Problem 2

$$5(3x - 2) = -2(x + 4)$$

$$15x - 10 = -2x - 8$$

$$17x = 2$$

$$x = \frac{2}{17}$$

### Problem 3

$$P = 2L + 2W$$

$$P - 2L = 2W$$

$$W = \frac{P - 2L}{2}$$

### Problem 4

$$-3x + 5 \leq -4$$

$$-3x \leq -9$$

$$x \geq 3$$

Interval notation:  $[3, \infty)$

Graph: Closed circle at 3, arrow pointing right

### Problem 5

$$|2x - 3| < 7$$

$$-7 < 2x - 3 < 7$$

$$-4 < 2x < 10$$

$$-2 < x < 5$$

Interval notation:  $(-2, 5)$



**Problem 6**

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3 - 5}{6 - 2} = \frac{-8}{4} = -2$$

**Problem 7**

$$y = \frac{3}{4}x - 2$$

**Problem 8**

Perpendicular slope:  $\frac{1}{2}$

Using point-slope form:

$$y - (-1) = \frac{1}{2}(x - 3)$$

$$y + 1 = \frac{1}{2}x - \frac{3}{2}$$

$$y = \frac{1}{2}x - \frac{5}{2}$$

**Problem 9**

$$3x - 4 = -x + 8$$

$$4x = 12$$

$$x = 3$$

$$y = 3(3) - 4 = 5$$

Solution:  $(3, 5)$

**Problem 10**

Equation:  $C = 100 + 50m$

After 6 months:  $C = 100 + 50(6) = 100 + 300 = \$400$

**Problem 11**

Let  $n$  = notebooks,  $p$  = pens

$$n + p = 80$$

$$3n + 2p = 190$$

From first equation:  $p = 80 - n$

Substitute:  $3n + 2(80 - n) = 190$

$$3n + 160 - 2n = 190$$

$$n = 30$$

Answer: 30 notebooks

**Problem 12**

Combined speed:  $55 + 65 = 120$  mph

Distance = Rate  $\times$  Time

$$300 = 120t$$

$$t = 2.5 \text{ hours}$$